

Boundary Case: Simultaneous Lifecycle Events

Governance Implications When Multiple Births, Deaths, and Matings Occur Concurrently, Testing Timestamp Ordering, Governance Sequencing Requirements, and Lineage Chain Integrity Under Concurrent Event Handling

Derivation Note B6.12 — CKS Series B, Phase 6 (Boundary Cases)

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, in operational terms, the governance sequencing requirements that arise when multiple lifecycle events — births, matings, and deaths — occur simultaneously or in rapid succession within a single governance-authorized restructuring window.

Abstract

The Coordination Knowledge Substrate (CKS) architecture treats birth, mating, and death as governed primitives, each producing an individual provenance record that must carry a distinct, correctly ordered timestamp. Most deployments exercise these primitives sequentially, and the provenance record for each event anchors without ambiguity to the event before and after it. The simultaneous lifecycle events boundary case tests what the architecture requires when this serial assumption fails: when a governance-authorized restructuring involves five cell births, three cell deaths, and one mating event producing two offspring, all occurring within a short governance window. The boundary is threefold. First, timestamp ordering must remain unambiguous even when underlying system clock resolution is sub-second — concurrent events cannot share a timestamp and still satisfy path retraceability. Second, lineage chain integrity requires a specific sequencing of record creation when mating and death are concurrent: offspring birth records must be created before parent death records, so the cross-lineage references the offspring carry anchor to living parent birth records rather than to entities already marked closed. Third, when the entire restructuring is authorized by a single governance decision, that batch authorization does not reduce per-event provenance to a single record — each lifecycle event must still produce its own individual entry. This note formalizes the governance sequencing protocol that concurrent lifecycle events require, states the stress points at which the architecture is most vulnerable under concurrency, and identifies the architectural limits at which governance must supply protocols the architecture does not itself specify.

1. Configuration description

The configuration under examination is a governance-authorized deployment restructuring in which the following lifecycle events occur within a short time window — too short for serial governance treatment, but within a single authorized governance decision:

Five new cells born, each with a distinct content-domain, created to cover capability areas the restructured deployment requires;

Three existing cells closed, each via functional obsolescence (Paper 2, §6.4) — their functions are no longer needed in the restructured deployment, and their substrate resources are to be released;

One mating event, combining two parent cells to produce two offspring cells that carry cross-lineage content from both parents.

All eleven lifecycle events (five births, three deaths, one mating record, two offspring births) are part of a single governance-authorized restructuring. The human authority that governs the deployment has reviewed and authorized the complete restructuring, and all eleven lifecycle events are expected to occur and be recorded before the restructuring window closes.

The configuration is representative of a class of real operational situations: enterprise deployments reorganized around new business functions, team structures redefined after a strategic shift, or AI Self architectures refactored as new capabilities absorb old ones. In each case, the governance decision to restructure is singular, but the lifecycle events it produces are multiple and nearly simultaneous.

2. Architectural boundary tested

Three boundaries are under test simultaneously, and they interact.

Boundary 1 — Timestamp ordering under concurrency. The CKS architecture requires that every lifecycle event produce an individual provenance record with distinct fields, including a timestamp. Path retraceability (Paper 1, §3.1; A1.07) requires that the causal and governance sequence of events be reconstructable from substrate content alone. When events occur nearly simultaneously, the timestamp field is the primary — and often sole — ordering signal. If two events receive the same timestamp, their ordering becomes ambiguous, and path retraceability fails for any query whose answer depends on which event preceded which. The boundary being tested is whether the governance and substrate machinery can guarantee distinct timestamps for events that occur within seconds or milliseconds of each other.

Boundary 2 — Lineage chain integrity when mating and death are concurrent. A mating event produces offspring whose provenance records carry cross-lineage references to the birth records of both parent cells. If either mating parent is subsequently closed — as in the restructuring configuration above, where both mating parents are among the cells being functionally replaced by the offspring and the five new cells — then the death record of a mating parent must be created after the offspring's birth record has been created and its cross-lineage references established. If parent death records are created first, the offspring's cross-lineage references point to entities whose death records exist, and the lineage chain carries a reference to a closed entity as a living parent — a contradiction the substrate cannot cleanly resolve and that path retraceability cannot explain.

Boundary 3 — Batch authorization producing individual records. The restructuring is authorized by a single governance decision. The question is whether that batch authorization is architecturally compatible with the per-event record requirement. The boundary tests whether governance can authorize a set of events as a unit while still producing an individual provenance record for each event within the set.

3. Governance implications

Governance sequencing protocol. Concurrent lifecycle events require governance to establish an explicit sequencing protocol — an ordered procedure for record creation — that the concurrent event handling must follow. The protocol does not need to slow the events themselves; it governs the order in which substrate records are written, which may be completed faster than human governance timescales and still satisfy the ordering requirement. The protocol for the restructuring configuration consists of four rules applied in order:

Sequencing Rule 1 — Mating records precede offspring birth records. A mating event must exist as a committed substrate record before any offspring from that mating can be born. Offspring cannot be born before the mating that creates them is recorded, because the offspring's birth record references the mating event and the parent birth records. A mating record written after an offspring birth record would create a forward reference the substrate could not anchor.

Sequencing Rule 2 — Offspring birth records precede parent death records when parent death follows mating. When a mating parent is subsequently closed, the offspring's birth record must be created and its cross-lineage references committed to substrate before the parent's death record is written. This preserves the integrity of the lineage chain: the offspring's reference to the parent's birth record resolves to an entity with no death record at the time the offspring's birth record is written, which is the correct structural state for a living parent.

Sequencing Rule 3 — Replacement entities are born before predecessors die. Where a cell death is of the functional-obsolescence type and the closed cell's function is taken up by a new cell, the replacement cell's birth record must be created before the predecessor's death record. Sequencing the birth before the death produces a substrate record that makes the replacement relationship explicit and retraceable.

Sequencing Rule 4 — Timestamps must reflect the governance sequencing order. Provenance records for the event sequence must carry timestamps that reflect Rules 1 through 3. Where system clock resolution makes natural timestamp ordering unreliable within the governance window, governance must impose timestamp discipline — explicitly assigning or confirming timestamps in the correct sequencing order rather than relying on wall-clock assignment.

Batch authorization with individual records. The restructuring may be authorized as a single governance decision — one authorization record in the substrate naming the complete restructuring, authorized by the responsible human authority, with the scope of the authorization covering all eleven lifecycle events. This approach is efficient and does not violate any architectural commitment. However, batch authorization scope does not substitute for per-event provenance. Each of the eleven lifecycle events must produce its own individual substrate record carrying the required provenance fields for that event. The batch authorization is a governance input to each event record — referenced in each event's provenance as the authorizing decision — but it is not a replacement for the event records themselves. A substrate that records the batch authorization without producing per-event records is not a substrate in which each lifecycle event can be individually traced,

audited, or interrogated.

Lineage consistency check. After the concurrent events complete, governance should execute a lineage consistency check over the restructuring window's event records. The check verifies: all five new births have complete birth records with anchors; all three functional-obsolescence deaths have complete death records with releasing provenance; the mating record exists and is committed; both offspring birth records exist and carry cross-lineage references that resolve to existing (not yet dead) parent entities; parent entities whose death records exist do not appear as living lineage anchors in any birth record created after their death record was committed.

4. Boundary tests

The following four tests operationalize the boundaries identified in §2.

Test (a) — Distinct timestamps in sequencing order. For every pair of lifecycle events in the restructuring window that have a governance sequencing relationship, the provenance timestamp of the earlier event must be strictly earlier than the provenance timestamp of the later event. No two events in the window may share an identical timestamp. A window that passes this test has substrate records from which the governance sequence can be reconstructed by reading timestamps alone.

Test (b) — Lineage chain consistency. Every birth record in the restructuring window must carry lineage anchors that resolve to substrate entities with no death records committed at or before the birth record's timestamp. No birth record may carry a reference to an entity whose death record timestamp precedes the birth record's timestamp. A window that passes this test has no dangling references and no orphaned offspring.

Test (c) — Offspring cross-lineage reference integrity. For the mating event producing two offspring, both offspring birth records must reference the mating event record and the birth records of both parent cells. Those references must resolve to committed substrate records. If either parent's death record exists, its timestamp must be later than the offspring's birth record timestamp. A window that passes this test confirms that the mating-death interaction was handled in the sequence Rule 2 requires.

Test (d) — Individual records under batch authorization. If the restructuring was authorized by a batch governance decision, the substrate must contain eleven distinct lifecycle event records — five birth records, three death records, one mating record, two offspring birth records — each carrying its own provenance fields including a distinct timestamp, a reference to the batch authorization decision, and the other provenance fields the architecture requires. A count of lifecycle event records in the restructuring window that falls below eleven means one or more events share a record or were absorbed into the batch authorization record, and per-event traceability has been lost.

5. Stress points

Stress point 1 — Timestamp collision. When multiple events are recorded within a short governance window by an automated substrate-writing process, system clock resolution may be insufficient to guarantee distinct timestamps for events occurring within the same millisecond or microsecond. If two events receive identical timestamps, their ordering becomes indeterminate from substrate content alone. This failure violates

path retraceability (A1.07): a reader of the substrate cannot determine which event preceded which, and any lineage chain whose integrity depends on the ordering of those two events becomes unresolvable. The risk is highest for the offspring birth records and for the records at the mating-death boundary. Governance must explicitly assign or confirm timestamps in sequencing order for events within the window rather than relying on wall-clock assignment alone.

Stress point 2 — Mating-death race condition. If the substrate-writing process for concurrent events is not sequencing-protocol-aware, the death records for mating parents may be written before the offspring birth records are committed — particularly if the three cells being closed are processed first while the mating and offspring machinery is still executing. If parent death records are committed before offspring birth records, then when the offspring birth records are subsequently written, they carry cross-lineage references to parent entities that already have death records. The lineage chain carries a reference to a closed entity as a living parent, a contradiction the substrate cannot cleanly express and that path retraceability cannot explain without reconstructing the race condition itself. Recovery from this failure requires re-sequencing records after the fact — a post-hoc correction that itself requires a governance record — and any audit of the restructuring window between the race condition and the correction will see an inconsistent lineage state. Sequencing Rule 2 is the preventive measure.

Stress point 3 — Batch authorization completeness failure. If the governance machinery treats the batch authorization decision as the authoritative substrate record for the entire restructuring, and produces no per-event records (or produces partial per-event records, covering only some of the eleven events), then the events not individually recorded are substrate-invisible. They may be inferable from state changes in the substrate — five new cells exist, three old cells are closed — but they are not individually traceable, their provenance fields are absent, and any query requiring the timestamp, authority, or antecedent chain for a specific event cannot be answered from substrate content alone. This failure mode is distinct from the timestamp collision and race condition failures in that it arises from a governance process design decision rather than from system clock or concurrency mechanics. The preventive measure is an explicit governance requirement that batch authorization scope does not reduce per-event record granularity.

6. Architectural limits

The CKS architecture, as Paper 2 commits to it, specifies that lifecycle events each produce individual provenance records with the fields the path retraceability commitment requires. The architecture does not specify how concurrent lifecycle events are to be handled — it does not define a concurrency protocol, a sequencing enforcement mechanism, or a race condition prevention procedure. This is not a gap in the architecture; it is the correct scope boundary between architecture and governance. The architecture specifies what each event record must contain; governance establishes the sequencing rules, the timestamp discipline protocol, and the completeness verification procedure that concurrent events require.

The boundary case formalizes the governance sequencing requirements that concurrency produces without asserting that those requirements are architectural commitments of Paper 2. They are governance obligations derived from the interaction of Paper 2's per-event record requirement with the operational reality of concurrent events — obligations that any deployment handling concurrent lifecycle events must satisfy to preserve path retraceability and lineage chain integrity, but that governance must supply, not the architecture.

7. Summary

Simultaneous lifecycle events test the CKS architecture at three points: the timestamp ordering that path retraceability depends on, the lineage chain integrity that mating-and-death concurrency threatens, and the per-event record granularity that batch authorization can obscure. The governance response is a four-rule sequencing protocol (mating before offspring birth; offspring birth before parent death; replacement before predecessor; timestamps reflecting the sequence), an explicit requirement that batch authorization produce individual records, and a post-window lineage consistency check. The three stress points — timestamp collision, mating-death race condition, and batch authorization completeness failure — are each preventable by the governance protocol, and each represents a distinct failure mode that would compromise path retraceability in a different way. The architectural limits of this boundary case are the correct scope limits: the architecture specifies what event records must contain; governance specifies how concurrency is handled.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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